# ELECENG 2EI4

# Project 4: XOR Gate

Humza Masood

Apr. 5, 2026

## Table of Contents

*Schematic* ..... 3

*Ideal sizing* ..... 4

i. Ideal Ratio ..... 4

ii. Hardware implementation ..... 4

*Testing* ..... 4

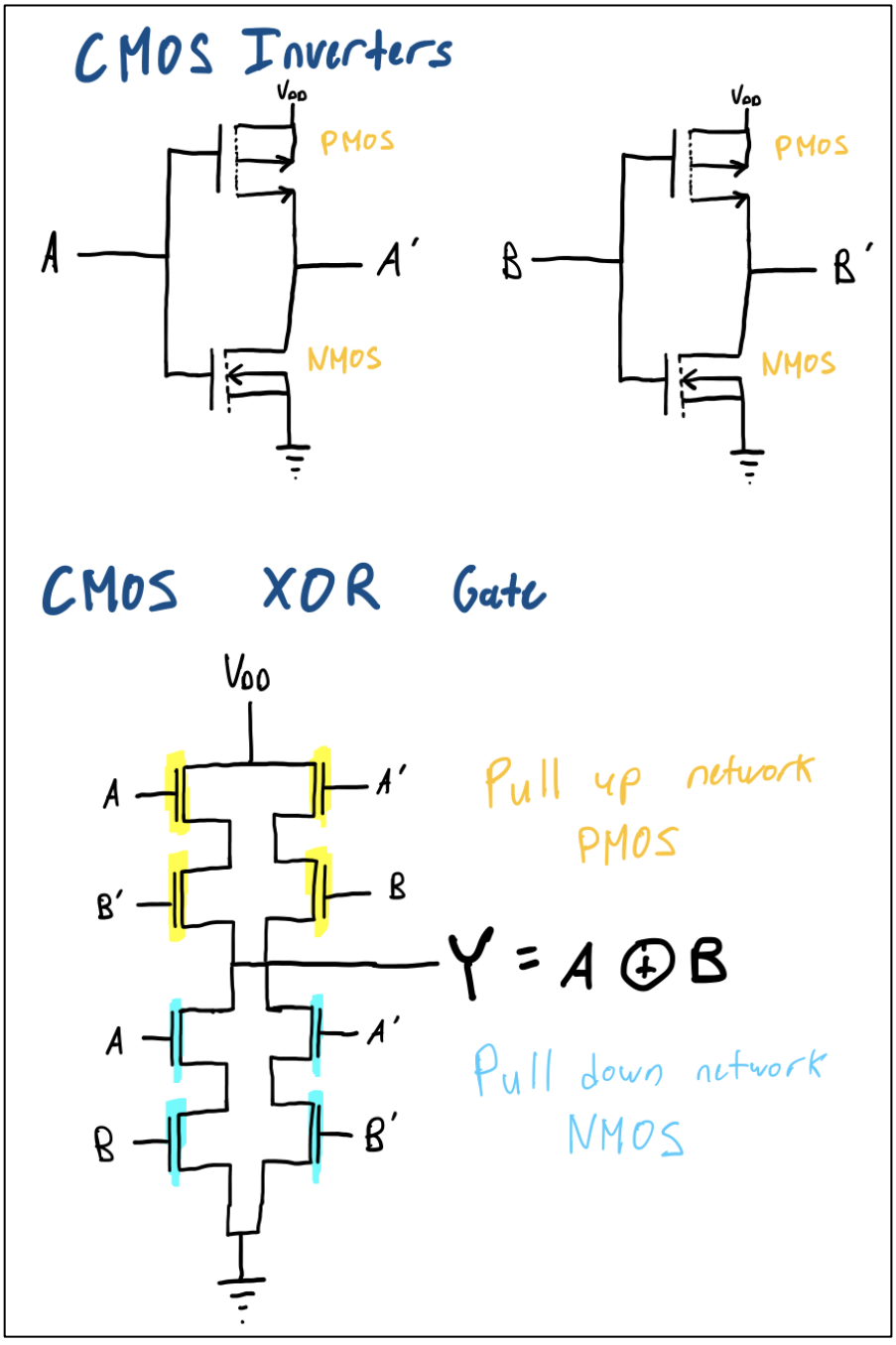
i. Physical circuit ..... 5

ii. Functional Testing ..... 6

iii. Static level testing ..... 7

*Timing* ..... 7

# Schematic




An XOR gate has the following Boolean expression: $Y = A \oplus B = \bar{A}B + A\bar{B}$. To implement this using a CMOS, the inverted equation was used: $\bar{Y} = AB + \bar{A}\,\bar{B}$. The pull-down network features two parallel branches, each with two NMOS in series, one for AB, and another for $\bar{A}\,\bar{B}$. The pull-up network allows a high output for $\bar{A}B$ or $A\bar{B}$. Two

CMOS inverters, each using an NMOS and a PMOS were used to obtain the inverted inputs.

# Ideal sizing

## i. Ideal Ratio

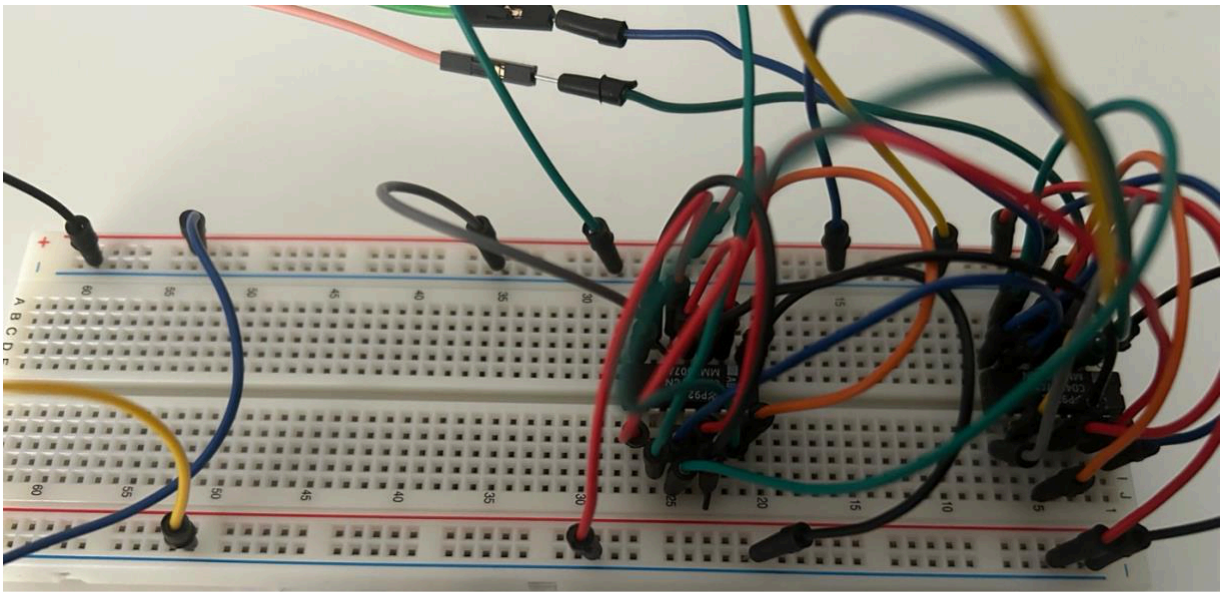
The pull-up and pull-down networks ideally have equal drive strength for the CMOS switching to be balanced. Therefore, the PMOS to NMOS width ratio is obtained using $\frac{\mu_n}{\mu_p}$, with the assumption that $W_p = 2W_n$. For this project, the transistors are assumed to have $(W/L)_n = 2/1$ and $(W/L)_p = 5/1$, meaning the sizing ratio is simply 2.5. Both the pull-up and pull-down network at worst will have two transistors in series, and since transistors in series increase the effective resistance, and ideally the overall path resistance is equal to the reference inverter, they should be twice as wide as the inverter transistor. Given the assumed (W/L) values of the inverter transistor, the XOR transistors should then have $(W/L)_n = 4/1$ and $(W/L)_p$ = 10/1.

## ii. Hardware implementation

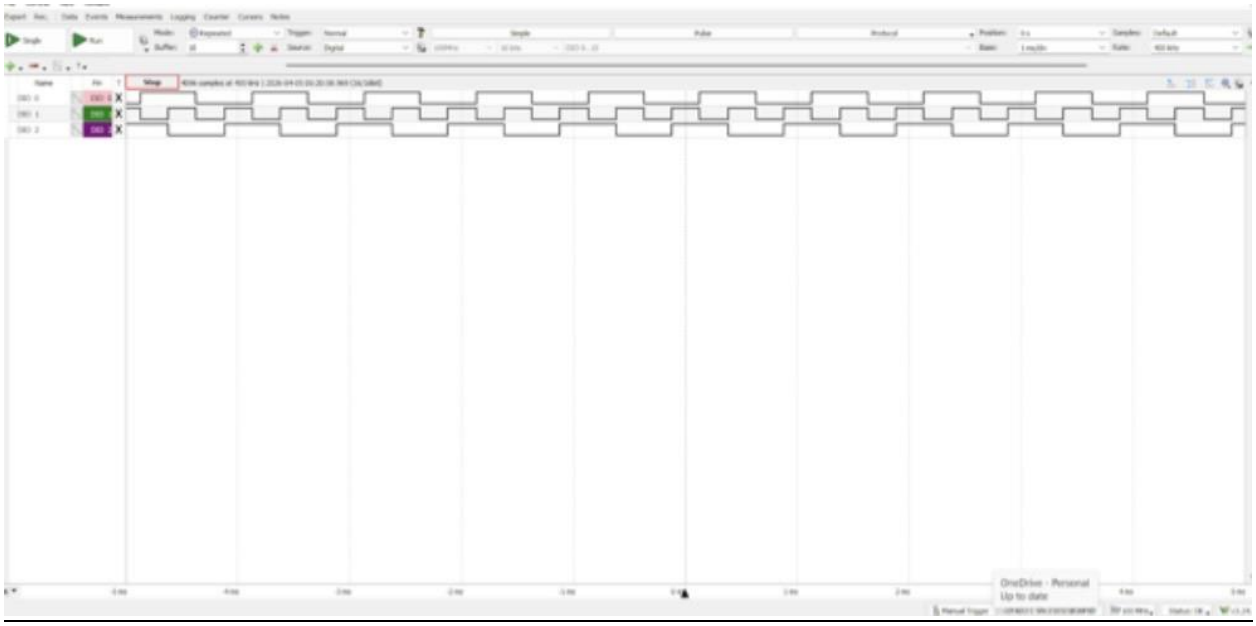
The IC provided for this course, the CD4007, does roughly preserve the PMOS to NMOS ratio, however the exact sizing is impossible to implement, as the dimensions of each MOSFET are chosen and designed from the manufacturer, meaning the user can not individually adjust them. The design should still behave correctly, but will likely have slightly worse rise and fall times than if they were ideal sized.

# Testing

i. Physical circuit



## ii. Functional Testing



For functional testing, DIO0 was used to drive input A and DIO1 was used to drive input B. The output, Y, was read by DIO2. All 4 combinations of inputs were tested, and the output was consistent with the truth table for an XOR gate, where the signal was high only when exactly one input was high.

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

### iii. Static level testing



For static level testing, input A was driven high, while input B was driven by a square wave that alternates between 0 and 5V. The output was observed, and since it alternates between low and high, the XOR gate is operating rail-to-rail under static operation. The measured $V_H$ was 4.97V and the measured $V_L$ was 0.01V. Even upon switching which input was constant and which was the square wave, the output remained the same, since the network is symmetric.

## Timing

For timing analysis, input A was driven high, while input B was driven by a square wave that alternates between 0 and 5V. In addition, a 100nF capacitor was connected at the output to simulate a load. The rise and fall time of the output waveform were found to be 457µs and 490µs respectively. The low-to-high and high-to-low propagation delays were determined between the input and output waveforms, to be 176µs and 201µs respectively. From these values, the propagation delay was calculated, using $\tau_P = \frac{\tau_{PLH}+\tau_{PHL}}{2}$, to be 188.5µs.